

Shift Management System

Japanese version: [README.ja.md](#)

This project started as an internal shift submission tool and has grown into a production shift management system used by 9 stores and 130+ staff members.

The system helps store staff submit their availability, allows managers to review and manage shifts, and exports finalized schedules to Excel for store operations.

Overview

The project is built for real store operations, where managers need to handle staff availability, required staffing levels, business hours, holidays, and Excel output without breaking the existing workflow.

It currently supports:

- Staff shift submission
- Manager shift review
- Staff and period management
- Business-hour settings
- Required staff settings
- Shortage detection
- Excel export
- Owner / Manager / Staff role separation
- Auto Shift V1

Scale

- 9 stores
- 130+ staff users
- Owner / Manager / Staff roles

Tech Stack

- Next.js App Router
- TypeScript
- Supabase PostgreSQL
- Supabase RLS
- Vercel
- ExcelJS

Key Features

Store Operation Settings

Managers can configure business hours, required staffing levels, peak-time settings, and date-specific operational rules for each store.

Shift Submission

Staff members can submit their available working times for each scheduling period.

Manager Operations

Managers can review submitted availability, manage staff, configure periods, adjust business hours, check staffing shortages, and export finalized schedules to Excel.

Permission Design

The system separates users into Owner, Manager, and Staff roles.

Managers are only allowed to access data for their own store. Server-side APIs are used for privileged operations, and the Supabase service role key is never exposed to the client.

Excel Export

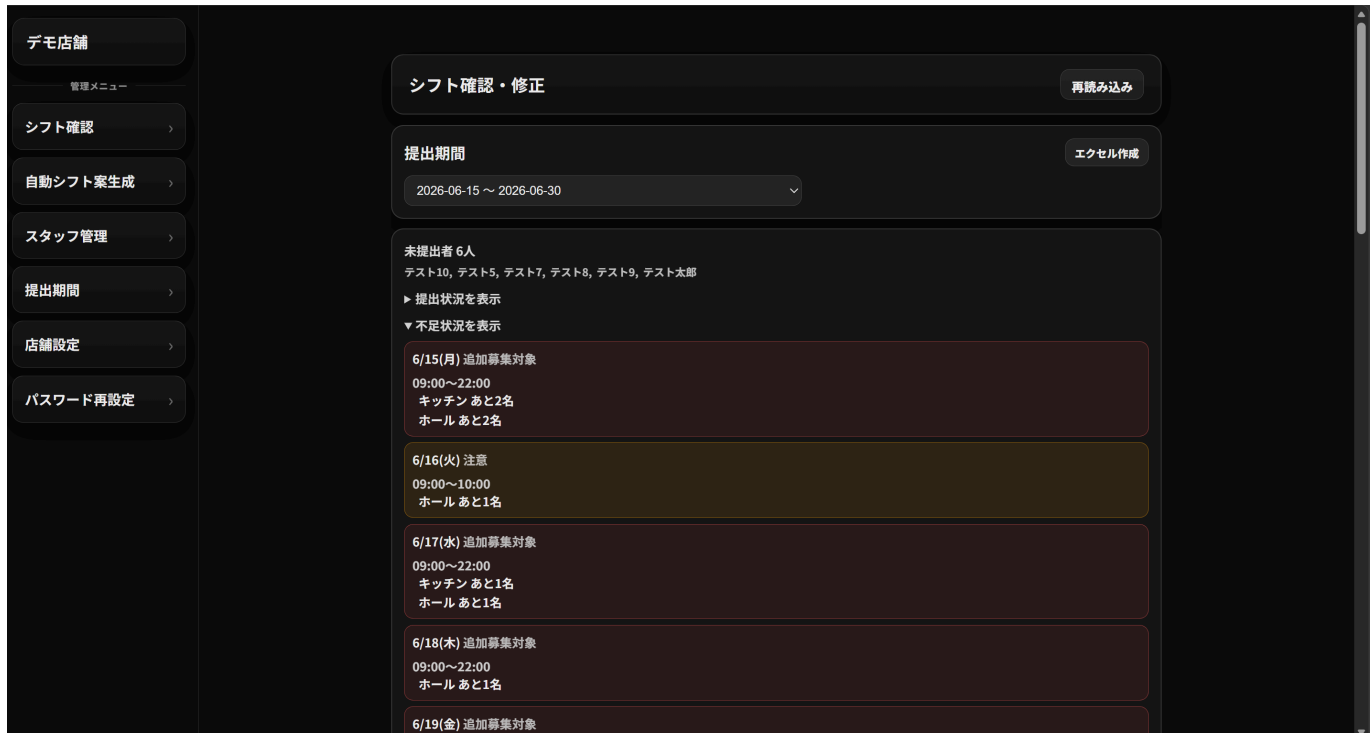
Finalized shift data can be exported to Excel, which is still an important part of the store operation workflow.

Screenshots

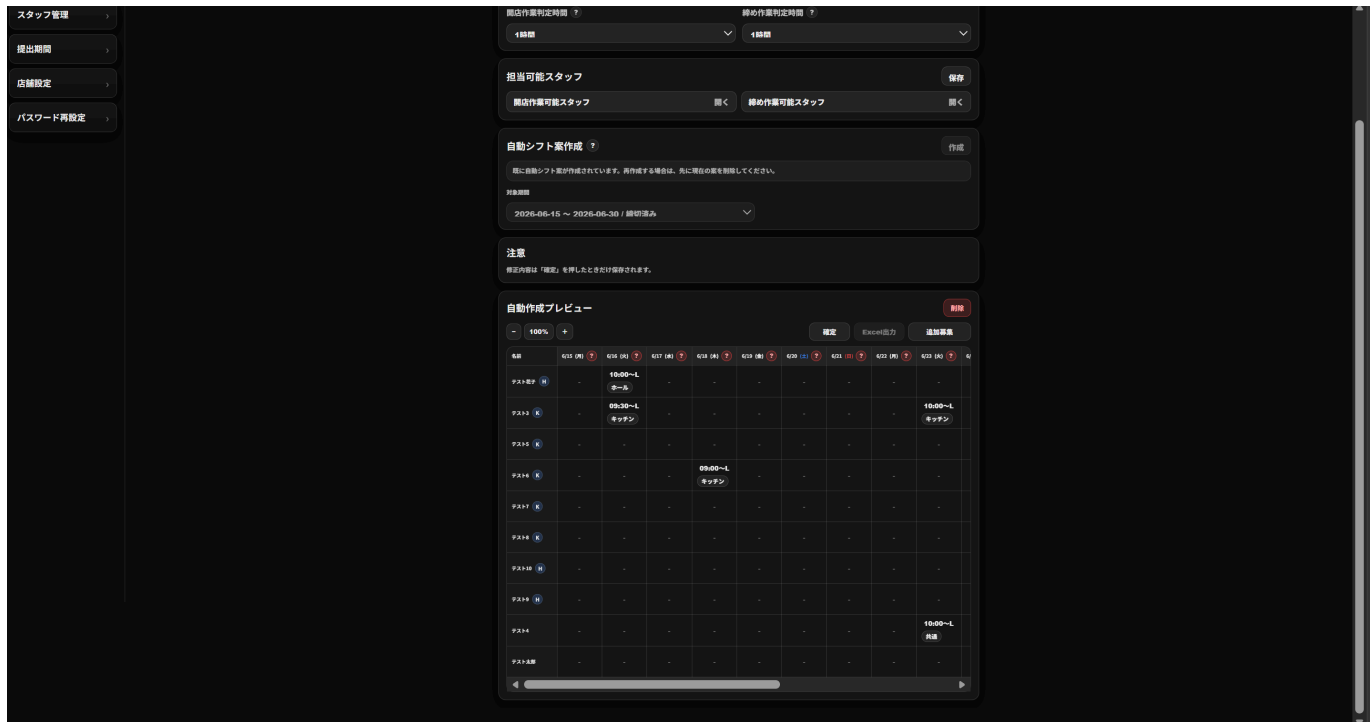
Manager Shift Calendar



Shortage Detection



Auto Shift Preview



Excel Export

[illegible]

デモ店舗

管理メニュー

シフト確認

自動シフト案生成

スタッフ管理

提出期間

店舗設定

パスワード再設定

スタッフ管理

再読み込み

スタッフ追加

名前
例：山田

コード
例：12

区分
共通

☐ 新人 ☒ 店長

追加

※ コードは2桁で設定してください。

全員 10 店長 1 キッチン 5 ホール 2 共通 2

スタッフ一覧

すべて保存済み 保存

順	名前	コード	属性	区分	操作
1	テスト花子	11	☐ 新人 ☒ 店長	ホール	削除
2	テスト太郎	11	☐ 新人 ☐ 店長	共通	削除
3	テスト3	11	☐ 新人 ☐ 店長	キッチン	削除
4	テスト4	11	☐ 新人 ☐ 店長	共通	削除

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ラベル	期間	状態	操作
-	2026-06-15 ~ 2026-06-30	有効中	有効中 削除
-	2026-06-12 ~ 2026-07-21	未適用	有効化 削除
-	2026-05-29 ~ 2026-07-01	未適用	有効化 削除
-	2026-05-29 ~ 2026-06-17	未適用	有効化 削除

Store Operation Settings

日付	営業時間	削除不可
2026-04-23	09:00 ~ 22:00	削除不可
2026-05-07	09:00 ~ 13:00	削除不可
2026-06-04	09:00 ~ 20:00	削除不可

Auto Shift V1

Auto Shift V1 is a rule-based draft generation feature for store managers.

The goal is not to fully automate scheduling, but to generate an initial draft that managers can review and adjust.

To avoid damaging real operation data, the system separates each stage of the shift data:

- **availabilities**: original staff submissions
- **draftRows**: auto-generated draft

- `workingRows`: manager-edited temporary state in the UI
- `confirmedRows`: manager-confirmed final schedule
- `confirmed_shift_snapshots`: final snapshot saved at confirmation time

This design keeps the original submitted availability data untouched while making it possible to compare generated drafts with manager-confirmed results later.

See: [Auto Shift V1 Case Study](#)

Database Design

The database is organized around stores, staff, periods, availability submissions, business-hour settings, required staffing rules, and Auto Shift V1 draft/confirmation records.

See: [Database Schema Notes](#)

Operational Considerations

Because this system is used in real stores, stability is more important than adding features quickly.

Important constraints:

- Do not overwrite original staff availability data
- Keep existing manual Excel export behavior working
- Prevent managers from accessing other stores' data
- Keep service role access server-side only
- Store generated drafts and confirmed schedules separately
- Preserve confirmed snapshots for future analysis

Future Work

- Analyze manager edits between generated drafts and confirmed schedules
- Add row-level identifiers for more accurate edit tracking
- Collect manager edit reasons
- Improve Auto Shift V2 using real operation data
- Add date-specific required staff overrides